

Question Paper

Exam Date & Time: 19-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: ECE 1071/ECE_1071 BASIC ELECTRONICS

Marks: 50

Duration: 180 mins.

Answer all the questions.

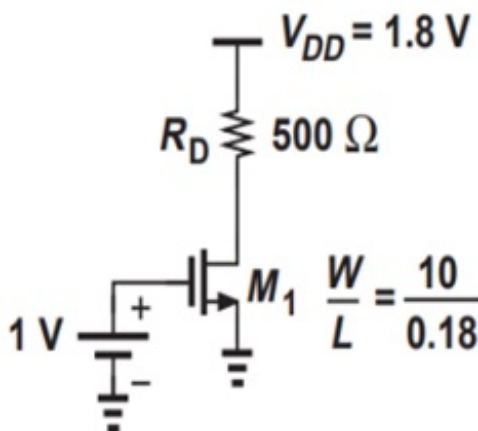
1A) A silicon diode has reverse saturation current of 1pA at 20°C . (4)

Determine:

- i) Diode bias voltage when $I_D = 3\text{mA}$.
- ii) Diode current when temperature is 100°C .

Course Outcome: "CO1" / Cognitive Level: "Analysis"

1B) What is the minimum allowable value of V_{DD} if M_1 must not enter the triode region if $\mu_n C_{ox} = 200\mu\text{A}/\text{V}^2$ (3)



Course Outcome: "CO2" / Cognitive Level: "Application"

1C) With appropriate circuit and waveforms, explain the working of half wave rectifier with capacitive filter. (3)

Course Outcome: "CO2" / Cognitive Level: "Understand"

2A) A Half wave rectifier is given an input of rms 220V , 50Hz through a transformer with turns ratio $10:1$. If the circuit uses an ideal diode, determine (a) DC output voltage (b) Rectifier efficiency (c) DC output voltage if a capacitor of 1mF is connected across $1\text{k}\Omega$ load resistance. (4)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

2B) For a Zener network, $R_S = 500\Omega$, $V_Z = 12\text{V}$ and $V_i = 50\text{V}$. Determine the minimum and maximum values of R_L and load current so that Zener diode remains in ON state and safe, given that $I_{Z\text{Max}} =$ (3)

2 mA and $P_{ZMin} = 500\text{mW}$. Draw the circuit diagram for the same.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 2C) Explain the significance of CMRR in an op-amp. If inputs $V_1 = 60\mu\text{V}$ and $V_2 = 30\mu\text{V}$ is given to the non-inverting and inverting terminals of an OPAMP, respectively. The common mode gain of the OPAMP is 40dB and CMRR is 60dB. Calculate the output voltage. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3A) Implement the following output function using inverting op-amps. Choose feedback resistance as $R_f = 10\text{k}\Omega$. (4)
 $V_o = V_1 - 5V_2 - 3V_3$

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3B) A 7-bit even parity Hamming code is used to encode a given 4-bit data. The codeword received at the receiver is "0110101". Decode the codeword to determine the transmitted data, assuming a single error occurred. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3C) Calculate using 2's complement subtraction: (3)
(i) $68.75_{(10)} - 27.50_{(10)}$
Convert $A23.35_{(16)}$ into Octal and Decimal equivalent.

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 4A) (i) Using KMAP, simplify the following Boolean expression and obtain the circuit? (4)

$$F = \sum_m (6, 7, 14, 15) + d(4, 5, 12, 13)$$

(ii) A Boolean expression has highest min-term of 35. How many cells are required in KMAP to express this problem? Justify your answer.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4B) With aid of relevant logical circuit, truth table and timing diagram Illustrate the working of negative edge triggered 4-bit ripple up counter using JK Flip flop (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4C) Analyze the operation of a 4-bit Serial-In Serial-Out (SISO) shift register by constructing its logical circuit and truth table for input data sequence of 1100. Explain how data propagates through the flip-flops with each clock pulse. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 5A) An audio signal of $15 \sin(2\pi \cdot 1500 \cdot t)$ volts amplitude is used to modulate a carrier of $60 \sin(2\pi \cdot 20000 \cdot t)$ volts. Calculate: (4)
i) Modulation index
ii) Sideband frequencies
iii) Bandwidth
Total power delivered if $R = 2\text{k}\Omega$

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- 5B) Demonstrate the working of a basic communication system with its block diagram. Explain each component. (3)

- 5C) Demonstrate the differences between ASK, FSK, and PSK by sketching and interpreting their waveforms with an example of 4-bit data. (3)

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_1071:BASIC
ELECTRONICS

#	Course Outcome	Description	Marks
1	1	CO 1	4
2	2	CO 2	16
3	3	CO 3	17
4	4	CO 4	13
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_1071:BASIC
ELECTRONICS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	6(12%)
3	Application	4	13(26%)
4	Analysis	6	21(42%)
5	Create	3	10(20%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)